the ill-understood laws of quantum gravity (Chapter 26). The foam is probabilistic in the sense that, at any moment, there is a certain probability the foam has one form and also a probability that it has another form, and these probabilities are continually changing. And the foam is truly tiny: the typical length of a wormhole would be the so-called Planck length, 0.00000000000000000000000000000001 centimeters; a hundredth of a billionth of a billionth the size of the nucleus of an atom. That's small!!

Back in the 1950s John Wheeler gave persuasive arguments for quantum foam, but there is now evidence that the laws of quantum gravity might suppress the foam and might even prevent it from arising.

If quantum foam *does* exist, I hope there is a natural process by which some of its wormholes can spontaneously grow to human size or bigger, and even did so during the extremely rapid "inflationary" expansion of the universe, when the universe was very, very young. However, we physicists have no hint of any evidence at all that such natural enlargement can or did occur.

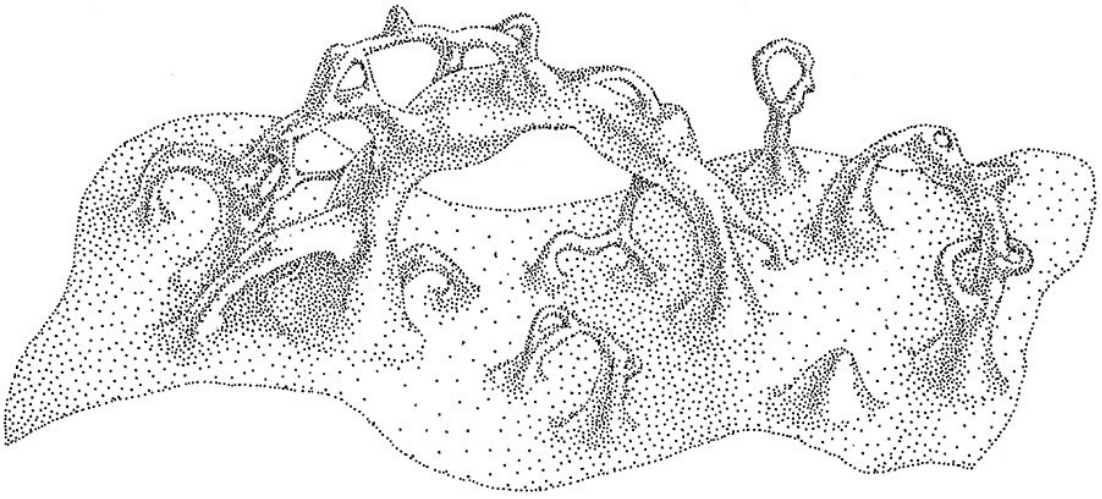


Fig. 14.7. Quantum foam. *[Drawing by Matt Zimet based on a sketch by me; from my book* Black Holes & Time Warps: Einstein's Outrageous Legacy.*]*